Python before pandas: 20 lines

PYTHON BASICS

Lists and loops

Your column of values

Data: sales = [1200, 800, 1500, 600, 950]

1 **Total of a list** → 5050

```
sum(sales)
```

2 **First and last value** → (1200, 950)

```
sales[0], sales[-1]
```

3 **Sort high to low** → [1500, 1200, 950, 800, 600]

```
sorted(sales, reverse=True)
```

4 **Keep values above 900** → [1200, 1500, 950]

```
[s for s in sales if s > 900]
```

5 **Add 18% GST to each** → [1416, 944, 1770, 708, 1121]

```
[round(s * 1.18) for s in sales]
```

Interview trap: Python counts from 0. sales[0] is the first value, sales[-1] the last.

PYTHON BASICS

Dictionaries

Lookup tables, like VLOOKUP

Data: region = {"Pune": "West", "Indore": "Central", "Kanpur": "North"}

1 **Look up a value** → West

```
region["Pune"]
```

2 **Safe lookup with a default** → Unknown

```
region.get("Delhi", "Unknown")
```

3 **Add a key, then count keys** → 4

```
region["Jaipur"] = "North"
```

4 **Count items in a list** → {'Pune': 2, 'Indore': 1}

```
from collections import Counter
city_count = Counter(["Pune", "Pune", "Indore"])
```

5 **Loop over key and value** → Pune:West, Indore:Central, ...

```
", ".join(f"{k}:{v}" for k, v in region.items())
```

Remember: Use .get() when a key may be missing. Square brackets crash with KeyError.

PYTHON BASICS

Strings that clean data

Names and codes from messy forms

Data: name = " ravi KUMAR "

1 Remove spaces → ravi KUMAR

```
name.strip()
```

2 Fix the case → Ravi Kumar

```
name.strip().title()
```

3 Split into first and last → [ravi, KUMAR]

```
name.strip().split()
```

4 Replace text → INV2024001

```
"INV-2024-001".replace("-", "")
```

5 Format money → Rs 1,234,567

```
f"Rs {1234567:,}"
```

Interview trap: Strings never change in place. name.strip() returns a new string; save it.

PYTHON BASICS · 1/2

Dates and errors

The two things that break scripts

Data: from datetime import date, datetime

1 Days between two dates

→ 78

```
(date(2024, 6, 20) - date(2024, 4, 3)).days
```

2 Text to a date

→ 2024-04-03

```
datetime.strptime("03-04-2024", "%d-%m-
%Y").date()
```

3 Date to month name

→ Apr 2024

```
date(2024, 4, 3).strftime("%b %Y")
```

PYTHON BASICS · 2/2

Dates and errors

The two things that break scripts

4 Write your own function

→ 1180

```
def gst(x, rate=0.18):
    return round(x * (1 + rate), 2)
```

5 Handle a bad value

→ 0

```
def to_num(x):
    try:
        return float(x)
    except ValueError:
        return 0.0
```

Say this: try/except keeps one bad row from stopping the whole report.

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